

MAT 319 HW04

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Problem 12.1. Let (s_n) and (t_n) be sequences and suppose there exists N_0 such that $s_n \leq t_n$ for all $n > N_0$. Show $\liminf s_n \leq \liminf t_n$ and $\limsup s_n \leq \limsup t_n$.

Solution. Define the sets

$$S_N = \{s_n : n > N\}, \quad T_N = \{t_n : n > N\}.$$

Claim — For all $N > N_0$, $\inf S_N \leq \inf T_N$ and $\sup S_N \leq \sup T_N$.

Proof. We know that $\inf S_N \leq s_n$ for all $n > N$ by definition of $\inf$, so we have

$$\inf S_N \leq s_n \leq t_n$$

for all $n > N$ (noting $s_n \leq t_n$ whenever $N > N_0$), which means that $\inf S_N$ is a lower bound for T_N . Since $\inf T_N$ is the greatest lower bound for T_N , $\inf S_N$ is at most as large as $\inf T_N$, which is what we wanted.

The proof for $\sup S_N \leq \sup T_N$ is analogous, but here it is anyway for completeness (yes, I copy-pasted and just changed the appropriate words). We know that $\sup T_N \geq t_n$ for all $n > N$ by definition of $\sup$, so we have

$$\sup T_N \geq t_n \geq s_n$$

for all $n > N$ (noting $t_n \geq s_n$ whenever $N > N_0$), which means that $\sup T_N$ is an upper bound for S_N . Since $\sup S_N$ is the least upper bound for S_N , $\sup T_N$ is at least as large as $\sup S_N$, which is what we wanted. $\square$

Now, to finish, $\inf S_N \leq \inf T_N$ being eventually true implies that

$$\liminf s_n = \lim_{N \rightarrow \infty} \inf S_N \leq \lim_{N \rightarrow \infty} \inf T_N = \liminf t_n,$$

which gives $\liminf s_n \leq \liminf t_n$. Similarly, because $\sup S_N \leq \sup T_N$ eventually,

$$\limsup s_n = \lim_{N \rightarrow \infty} \sup S_N \leq \lim_{N \rightarrow \infty} \sup T_N = \limsup t_n,$$

which gives $\limsup s_n \leq \limsup t_n$. $\square$

Problem 12.3. Let (s_n) and (t_n) be the following sequences with period 4:

$$(s_n) = (0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, \dots)$$

$$(t_n) = (2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, \dots)$$

Find

(a) $\liminf s_n + \liminf t_n$,

(b) $\liminf(s_n + t_n)$,

(c) $\liminf s_n + \limsup t_n$,

(d) $\limsup(s_n + t_n)$,

(e) $\limsup s_n + \limsup t_n$,

(f) $\liminf(s_n t_n)$,

(g) $\limsup(s_n t_n)$

Solution. First, we compute

$$(s_n + t_n) = (2, 2, 3, 1, 2, 2, 3, 1, 2, 2, 3, 1, 2, \dots)$$

$$(s_n t_n) = (0, 1, 2, 0, 0, 1, 2, 0, 0, 1, 2, 0, 0, \dots),$$

noting that $(s_n + t_n)$ and $(s_n t_n)$ are both sequences with period 4 (namely $s_{n+4} = s_n$ and $t_{n+4} = t_n$ implies that $s_{n+4} + t_{n+4} = s_n + t_n$ and $s_{n+4} t_{n+4} = s_n t_n$ for all n).

The following claim about $\liminf$ and $\limsup$ of a periodic sequence will now completely solve the entire problem.

Claim — Let (x_n) be a periodic sequence with period p , meaning $x_n = x_{n+p}$ for all natural n . Then,

$$\limsup x_n = \max\{x_1, x_2, \dots, x_p\}, \quad \liminf x_n = \min\{x_1, x_2, \dots, x_p\}.$$

Proof. Define $X_N = \{x_n : n > N\}$ and $P = \{x_1, x_2, \dots, x_p\}$. Since each element of $\{x_n\}$ is equal to one of $x_1, \dots, x_p$, $X_N \subseteq P$. On the other hand, picking an integer q such that $pq > N$ (which exists by the archimedean property of $\mathbb{R}$) shows us that

$$P = \{x_{pq+1}, x_{pq+2}, \dots, x_{pq+p}\} \subseteq X_N,$$

which means we can conclude the equality of sets $X_N = P$. In particular,

$$\sup X_N = \sup P = \max P, \quad \inf X_N = \min P = \min P.$$

because $\{x_1, \dots, x_p\}$ is finite. Taking the limit then finishes:

$$\limsup x_n = \lim_{N \rightarrow \infty} \sup X_N = \lim_{N \rightarrow \infty} \max P = \max P,$$

$$\liminf x_n = \lim_{N \rightarrow \infty} \inf X_N = \lim_{N \rightarrow \infty} \min P = \min P. \quad \square$$

Now, (s_n) , (t_n) , $(s_n + t_n)$, and $(s_n t_n)$ are all periodic, so:

$$\liminf s_n = \min\{0, 1, 2, 1\} = 0, \quad \limsup s_n = \max\{0, 1, 2, 1\} = 2,$$

$$\liminf t_n = \min\{2, 1, 1, 0\} = 0, \quad \limsup t_n = \max\{2, 1, 1, 0\} = 2,$$

$$\liminf s_n + t_n = \min\{2, 2, 3, 1\} = 1, \quad \limsup s_n + t_n = \max\{2, 2, 3, 1\} = 3,$$

$$\liminf s_n t_n = \min\{0, 1, 2, 0\} = 0, \quad \limsup s_n t_n = \max\{0, 1, 2, 0\} = 2.$$

Then, the answers are:

(a) $\liminf s_n + \liminf t_n = 0 + 0 = 0$,

(b) $\liminf(s_n + t_n) = 1$,

(c) $\liminf s_n + \limsup t_n = 0 + 2 = 2$,

(d) $\limsup(s_n + t_n) = 3$,

(e) $\limsup s_n + \limsup t_n = 2 + 2 = 4$,

(f) $\liminf(s_n t_n) = 0$,

(g) $\limsup(s_n t_n) = 2$

□

Problem 12.6. Let (s_n) be a bounded sequence, and let k be a nonnegative real number.

(a) Prove $\limsup(ks_n) = k \cdot \limsup s_n$.

(b) Do the same for $\liminf$.

(c) What happens in (a) and (b) if $k < 0$?

Solution. Let S be the set of subsequential limits of (s_n) and kS be the set of subsequential limits of (ks_n) , the sequence obtained by multiplying each element of s_n by k . (We use kS to mean the set obtained by multiplying every element of S with k .) All parts of the question will rely on the fact that

$$kS = \{k \cdot \ell : \ell \in S\}.$$

This is obvious when $k = 0$ because (ks_{n_i}) is the zero sequence, and so $S_0 = \{0\}$. When $k \neq 0$, we rephrase the set equality above as follows:

Claim — The real number ℓ is in S if and only if $k \cdot \ell$ is in kS .

Proof. First, we show that, if ℓ is in S , then $k \cdot \ell$ is in kS . For any $\ell \in S$, there exists a subsequence (s_{n_i}) of (s_n) converging to ℓ . We know that

$$\lim_{i \rightarrow \infty} ks_{n_i} = k \lim_{i \rightarrow \infty} s_{n_i} = k \cdot \ell.$$

Obviously (ks_{n_i}) is a subsequence of the sequence (ks_n) ; to be more thorough, we can let $t_n = ks_n$ and then say that the t_{n_i} form a subsequence of (t_n) because (s_{n_i}) being a subsequence of (s_n) implies that n_i is an increasing function of i . Thus, $k \cdot \ell$ is a subsequential limit and therefore an element of kS .

Now, we prove the converse, namely that $k \cdot \ell \in kS$ implies $\ell \in S$. In fact, we can just reuse the previous part: Let $k' = \frac{1}{k}$ (noting that $k \neq 0$) and define $t_n = ks_n$, so that $s_n = \frac{t_n}{k} = k't_n$. If some $k \cdot \ell$ is a subsequential limit of (t_n) (and thus in kS), then we showed above that $k' \cdot (k \cdot \ell) = \ell$ must be a subsequential limit of the sequence $(k't_n) = (s_n)$, so $\ell \in S$. $\square$

Now, we can solve (a) and (b) by making use of the fact that $\limsup(ks_n) = \sup kS$ and $\liminf(ks_n) = \inf kS$, since, when $k \geq 0$,

$$\sup kS = k \sup S = k \cdot \limsup s_n, \quad \inf kS = k \inf S = k \cdot \liminf s_n.$$

For (c), we just have to recall that

$$\liminf(s_n) = -\limsup(-s_n), \quad \limsup s_n = -\liminf(-s_n),$$

as this lets us get

$$\liminf(ks_n) = -\limsup((-k)s_n) = -(-k) \cdot \limsup(s_n) = k \limsup s_n$$

for $\liminf$, and

$$\limsup(ks_n) = -\liminf((-k)s_n) = -(-k) \cdot \liminf(s_n) = k \liminf s_n$$

for $\limsup$. $\square$

Problem 12.8. Let (s_n) and (t_n) be bounded sequences of nonnegative numbers. Prove

$$\limsup s_n t_n \leq (\limsup s_n)(\limsup t_n).$$

Solution. Let $\bar{s}_N = \sup\{s_n : n > N\}$. Since the s_n and t_n are all nonnegative, $s_n \leq \bar{s}_N$ for $n > N$ implies that

$$s_n t_n \leq \bar{s}_N t_n$$

whenever $n > N$. This inequality is preserved when taking the $\limsup$ (see Problem 12.1; here $N_0 = N$) so

$$\limsup s_n t_n \leq \bar{s}_N \cdot \limsup t_n$$

for all N ; note that we can factor out $\bar{s}_N$ due to the previous problem (12.6). Then, we also take the limit as N goes to infinity to say that

$$\limsup s_n t_n = \lim_{N \rightarrow \infty} \limsup s_n t_n \leq \left(\lim_{N \rightarrow \infty} \bar{s}_N \right) (\limsup t_n) = (\limsup s_n)(\limsup t_n),$$

which is what we wanted to show. $\square$

Problem 14.6. (a) Prove that if $\sum |a_n|$ converges and (b_n) is bounded then $\sum a_n b_n$ converges.

(b) Observe that Corollary 14.7 is a special case of part (a).

Solution. For (a), it is equivalent to show that the sequence of partial sums of $a_n b_n$ is a Cauchy sequence. That is, for each $\varepsilon > 0$, we want to show the existence of an N such that $m, n > N$ implies that

$$\left| \sum_{k=m}^n a_k b_k \right| < \varepsilon.$$

Since (b_n) is bounded, there exists an M for which $|b_k| < M$ for all k . Thus, we can get the following bound via triangle inequality:

$$\left| \sum_{k=m}^n a_k b_k \right| \leq \sum_{k=m}^n |a_k b_k| < \sum_{k=m}^n M |a_k| = M \sum_{k=m}^n |a_k|.$$

Now, since $\sum |a_n|$ converges and $\frac{\varepsilon}{M}$ is a positive real, there exists an $\hat{N}$ such that, whenever $m, n > \hat{N}$,

$$\sum_{k=m}^n |a_k| \leq \frac{\varepsilon}{M},$$

which means that $m, n > \hat{N}$ implies

$$\left| \sum_{k=m}^n a_k b_k \right| < M \sum_{k=m}^n |a_k| < \frac{\varepsilon}{M} \cdot M = \varepsilon.$$

This is enough to show that the partial sums of $a_k b_k$ form a Cauchy sequence (taking $N = \hat{N}$), thus $\sum a_n b_n$ converges.

As for part (b), Corollary 14.7 (absolute convergence implies convergence) arises from taking $b_n = 1$ for all n . $\square$

Problem 14.7. Prove that if $\sum a_n$ is a convergent series of nonnegative numbers and $p > 1$, then $\sum a_n^p$ converges.

Solution. Since $\sum a_n$ converges, $\lim_{n \rightarrow \infty} a_n = 0$. This means that eventually $a_n < 1$, say for $n > N$ where N is an integer (to be precise, taking $\varepsilon = 1$, there exists an N such that $n > N$ implies $|a_n - 0| = a_n < \varepsilon$). For $n > N$, we can then say that $|a_n^p| = a_n^p < a_n$ so, by the comparison test,

$$\sum_{n=N+1}^{\infty} a_n^p \leq \sum_{n=N+1}^{\infty} a_n,$$

in particular the left-hand sum converges. This is equivalent to the statement

$$\text{for all } \varepsilon > 0, \text{ there exists a } B \text{ such that } n > m > B \implies \left| \sum_{k=m}^n a_k^p \right| < \varepsilon \quad (1)$$

by the Cauchy criterion for the series $\sum_{k>N} a_k^p$, where the sum is taken over all indices greater than N . On the other hand, this is also exactly the Cauchy criterion for $\sum a_k^p$ just taken over all natural numbers: that is, the chain of logic looks like

$$\sum_{k=N+1}^{\infty} a_k^p \text{ converges} \iff (1) \iff \sum_{k=1}^{\infty} a_k^p \text{ converges},$$

so $\sum a_k^p$ converges! □

Problem 14.9. For sequences (a_n) and (b_n) , suppose that the set $\{n \in \mathbb{N} : a_n \neq b_n\}$ is finite. Show that $\sum a_n$ converges if and only if $\sum b_n$ converges.

Solution. We show that $\sum a_n$ satisfies the Cauchy criterion if and only if $\sum b_n$ also does.

Let n_0 be an integer larger than every element of $\{n \in \mathbb{N} : a_n \neq b_n\}$, so that $k > n_0$ implies $a_k = b_k$. Then, $q > p > n_0$ implies that $a_k = b_k$ if $p \leq k \leq q$, hence $\sum_{k=p}^q a_k = \sum_{k=p}^q b_k$. From there we can see that

$$\text{for all } \varepsilon > 0, \text{ there exists an } N \text{ such that } q > p > \max\{N, n_0\} \implies \left| \sum_{k=p}^q a_k \right| < \varepsilon \quad (2)$$

is the exact same statement as

$$\text{for all } \varepsilon > 0, \text{ there exists an } N \text{ such that } q > p > \max\{N, n_0\} \implies \left| \sum_{k=p}^q b_k \right| < \varepsilon \quad (3)$$

The former statement is the Cauchy criterion for $\sum a_k$ and the latter is the Cauchy criterion for $\sum b_k$, so we have

$$\sum_{k=1}^{\infty} a_k \text{ converges} \iff (1) \iff (2) \iff \sum_{k=1}^{\infty} b_k \text{ converges},$$

which is what we wanted. □

Problem 14.14. Prove that $\sum \frac{1}{n}$ diverges by comparison with $\sum_{n \geq 2} a_n$ where (a_n) is the sequence

$$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}, \dots \right).$$

In general, $a_n = \frac{1}{2^r}$, where r is the integer satisfying $2^{r-1} < n \leq 2^r$ (and the sequence is indexed starting at 2 for some ridiculous reason).

Solution. We start by immediately correcting the horrendous indexing given to us: define the sequence (b_n) over all naturals such that $b_n = a_{n+1}$ for all n . Our goal is to prove the divergence of the harmonic series via comparison with $\sum b_n$.

Claim — Let (s_n) be the sequence of partial sums of (b_n) . Then, for every natural number k ,

$$s_{2^k-1} = \frac{k}{2}.$$

Proof. We induct on k . The base case $k = 1$ is trivial, as

$$s_1 = a_1 = \frac{1}{2}.$$

Supposing the claim holds for some k (so $s_{2^k-1} = \frac{k}{2}$), we can write

$$s_{2^{k+1}-1} = s_{2^k-1} + \sum_{j=2^k}^{2^{k+1}-1} a_j = \frac{k}{2} + \sum_{j=2^k}^{2^{k+1}-1} \frac{1}{2^{k+1}} = \frac{k}{2} + 2^k \cdot \frac{1}{2^{k+1}} = \frac{k}{2} + \frac{1}{2} = \frac{k+1}{2},$$

noting that $a_j = \frac{1}{2^{k+1}}$ when $2^k \leq j < 2^{k+1}$ because the smallest power of two larger than j is 2^{k+1} . Thus, the claim is proven by the principle of mathematical induction. $\square$

Now, since each b_n is positive, $s_{n+1} - s_n = a_{n+1} > 0$ means that (s_n) is monotonically increasing, so $\lim_{n \rightarrow \infty} s_n$ exists. Moreover, any subsequence of (s_n) converges to this same limit. Taking $n_k = 2^k - 1$, we look at the limit of the subsequence of (s_n) given by (s_{n_k}) :

$$\sum_{n=1}^{\infty} b_n = \lim_{n \rightarrow \infty} s_n = \lim_{k \rightarrow \infty} s_{n_k} = \lim_{k \rightarrow \infty} \frac{k}{2} = +\infty.$$

Thus, $\sum b_n$ diverges to $+\infty$.

Now, by definition, for any natural number n , we know that $a_{n+1} = \frac{1}{2^r}$ where r is an integer such that $2^{r-1} < n+1 \leq 2^r$. Thus, $n < 2^r$ implies

$$\frac{1}{n} > \frac{1}{2^r} = a_{n+1} = b_n.$$

Since (b_n) is a sequence of nonnegative numbers and $\frac{1}{n} > b_n$ for all n , the comparison theorem tells us that $\sum b_n$ diverging to $+\infty$ means that $\sum \frac{1}{n}$ also diverges to $+\infty$. $\square$